EARTH'S FIELD NMR — a proton precession spectrometer

A coil round a bottle of water. POLARIZE: a few amperes through the coil for a few seconds magnetize the water's protons along the coil's field, a few hundred times more than the Earth's 46 uT alone would. Switch off: in a millisecond the field is gone, faster than the protons can follow, and they precess about the Earth's field at f = 42.58 Hz/uT x B: 1960 Hz here, a few microvolts in the same coil, dying away in a fraction of a second to a few seconds (T2*). LISTEN: the relay hands the coil to a tank tuned to f, and the preamplifier and the filters raise the microvolts to a volt.
The reference coil (handbook): 916 turns of AWG 20 on 3-inch pipe, 27 mH, 8.9 ohm, Q 37. At 1.5 A: 7.8 mT, about 1.2 uV in the coil from 1 L of water, 45 uV across the tank, 0.9 V at OUT on HIGH gain.

POLARIZING SUPPLY

6–30 V, up to 3 A. A bench supply in constant current sets the coil's current: or the coil's resistance does.

V POL: behind the 3 A fuse

F1 3A — J2 — D5 S534, D6 SMBJ33A, C5 470uF — V POL — R3 15k — LED3 GREEN

LED3: the supply is there

COIL

A (2), B (1): the coil's two leads
S (3): its shield, if any
up to 130 V at switch-off:
do not touch
10–200 mH, up to 3 A, 1/2 L i^2 = 1 J

K1 HFD3/12: released, the coil LISTENS (as drawn);
energized, it is POLARIZED: A to V POL, B to the switch

J3 COIL A / COIL B — K1B HFD3/12 — V POL

TANK+ — TANK−

D8 SMBJ12CA

D9: a clamp, idle unless a fault dumps the coil's energy here

D7 SMCJ100A holds the drain near 130 V while the coil's current falls to zero in a straight line (0.4 ms for the reference coil): faster than the protons can follow

DRAIN — DRAIN: up to 160 V, a ×10 probe

Q1 IRF54227 — R5 47R, R6 100k — SOURCE — I POL — R4 0.1R

I POL: 0.1 V per ampere, on and ramping off

GATE

CAL IN

a generator through 1M: 1 = V / 1M
into the tank. mV-level; 5 Vpp max

J4 — R14 1M — R13 51R

SW1 TUNE

SW1 TUNE: each switch adds its capacitance (C0G) to the tank. In 3.3 nF steps up to 841 nF.
Resonance: f = 1 / (2 pi sqrt(L C)): tune to the Larmor frequency for the most signal.
The reference coil needs 246 nF at 1960 Hz. Caps C10–C29: 3.3n, 6.8n, 10n 0805, 47n, 100n 1206

gain 101 = 1 + 2 × 10k / 200R (between the two − inputs)

PREAMPLIFIER

an instrumentation amplifier, gain 101: U1 and U2 amplify each side of the tank, U3 takes the difference, so pickup common to both cancels. OPA197: 5.5 nV/rtHz and no current noise against the tank's 12 k: 1.7 dB noise figure with the reference coil.
R11, R12 (1M) give the inputs their DC: R15, R16 (1k) limit a fault's current into them

R11 1M, R15 1k — U1 1 — INA A — R17 10k — −15V
R19 200R
R18 10k — −15V — U2 2 — INA B — R16 1k, R12 1M — +15V

HIGH-PASS

4th-order Butterworth, 688 Hz: two unity-gain Sallen-Key sections,
Q 0.55 at 726 Hz and Q 1.31 at 670 Hz. 60 Hz −85 dB, 180 Hz −47 dB,
420 Hz −17 dB: mains hum and its low harmonics go. C0G capacitors.

R22 10k, R21 10k, R20 10k — U3 1 — R23 10k — PREAMP
C30 100nF, C31 100nF — R24 20k, R25 24k — U4 1 — HP1
C32 100nF, C33 100nF — R26 9.1k, R27 62k — U4 1 — HP

LOW-PASS

2nd-order Butterworth, 4097 Hz (Q 0.72):
the top of the band, and the anti-alias filter
for a sound card. C0G capacitors.

R28 8.2k, R29 8.2k — C34 6.8nF, C35 3.3nF — U4 1 — LP

GAIN

×21 or ×201 = 1 + 200k / Rg

U5 1 — R31 10k, R32 1k — SW2 GAIN
1: LOW, ×21.21 overall (67 dB)
3: HIGH, ×201.01 (96 dB)

U6 2 — R30 200k — OUT 1k — J5

OUT

the FID, about 1 V on HIGH:
1k in series: for a scope, an ADC
or a sound card's line input

POWER

+15 V, GND, −15 V,
100 mA a rail

J1 — D1 S534 — +15V — PWR_FLAG
D3 S534, C1 100nF, C3 100nF — R1 10k — LED1 GREEN
D4 S534, C2 100nF, C4 100nF — LED2 GREEN — R2 10k
D2 S534 — −15V — PWR_FLAG

D1, D2: a reversed supply does nothing
D3, D4: clamp a rail left open
LED1, LED2: both rails present

SEQUENCER

POL CMD high = POLARIZE; the relay pulls in at once, and the switch closes 18 ms later (U9).
POL CMD low: the switch opens at once, the coil's current ramps off through D7, and the relay lets go 15 ms later (U10): LISTEN.

RV1 POLARIZE TIME
0.2–4.7 s, clockwise longer

RV2 LISTEN TIME
0.2–4.8 s, clockwise longer

R40 100k, RV1 2M, RV2 2M — D10 1N4148W — R41 100k — D11 1N4148W — TIMER
C53 1u — U7 NE555 (RST, DISCH, CONT, THRES, TRIG, OUT, VCC, GND)

C50–C52: 3 × 1 uF polypropylene
film; t = 0.75 R C each way

C50 1u, C51 1u, C52 1u

SW3 SEQUENCE
1: EXT, the POLARIZE IN jack
3: INT, the timer (RV1, RV2)

SW4 MODE
1: RUN
3: LISTEN ONLY (no polarizing)

POL CMD — R42 100k

POLARIZE IN

high (> 1.3 V) = polarize: TTL, 3.3 V, 5 V,
a function generator: +/−30 V max;
open = listen

J6 — R43 100k — R44 10k — U8 1 — R48 10k — EXT
D12 BAV99 — R47 1M
R45 100k, R46 10k

U8: a comparator at 1.3 V (R45, R46),
147 mV of hysteresis (R47)

R49 DELAY 220k — D13 1N4148W — C55 100nF — U9 1 — GATE
R7 4.7k

U9 GATE: on 18 ms after POL CMD rises (R49, C55),
off at once when it falls (D13)

R50 HOLD 270k — D14 1N4148W — C56 100nF — U10 1 — RELAY
R9 10k

U10 RELAY: on at once (D14),
off 15 ms after POL CMD falls (R50, C56)

K1 coil
≈1120 ohm

D8 1N4148W — R10 3.3k — LED4 RED

red: POLARIZING

R8 10k — Q2 MMBT3904

R51 10k, R52 10k — 7.3 V — C54 100nF

R53 10k — Q3 MMBT3904 — R55 1k — J8 — R54 10k

SYNC OUT

follows POL CMD: 0 / 14 V through 1k:
its falling edge is the switch-off:
trigger the scope on it

GND — ground loops for the scope's clip

MH1 MH2 MH3 MH4

M3 mounting holes, one per corner

Bypass: 100 nF X7R at every op-amp's, comparator's and the timer's supply pins (C80–C97)

Open source, CERN-OHL-P-2.0 — github.com/gallicchio/earth-nmr-pcb
OPA197 for every amplifier, TS391 comparators, NE555 timer, IRFS4227 switch, HFD3 relay
Larmor frequency 1960 Hz at Harvey Mudd College, Claremont CA (46.0 uT, IGRF-14, 2026-11-01): 1

Jason Gallicchio and Claude

Sheets: /
File: earthnmr.kicad_sch

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